

Hallucination Detection Report

Generated: 2026-04-15 17:24 · Extracted triples: 7 · Verdicts: 4 · Inconsistencies: 0 · Symbolic violations: 3

Executive Summary

2 Correct	2 Fabricated Fact	
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Extracted Triples

All verifiable triples extracted from the LLM response before any canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
575	HAS_INTEREST_RATE	12.21%	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject.
580	HAS_INTEREST_RATE	11.91%	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject.
579	HAS_INTEREST_RATE	11.91%	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject.
579	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Specifies the risk tier classification of the subject.
890	HAS_INTEREST_RATE	Below 4.36%	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate associated with the subject, below a specified value.
890	HAS_RISK_TIER	super prime	Yes	ENTITY_ATTRIBUTE	Specifies the risk tier classification of the subject.
890	HAS_DOWN_PAYMENT	Below 2.0%	Yes	ENTITY_ATTRIBUTE	Indicates the down payment percentage associated with the subject, below a specified value.

Canonicalized Triples

Triples after predicate normalization against the database schema. Shows the original predicate, the matched canonical predicate, and match confidence.

Subject	Original Pred.	Object	Canonical Pred.	Confidence
575	HAS_INTEREST_RATE	12.21%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_INTEREST_RATE	Below 4.36%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
890	HAS_RISK_TIER	super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	MEDIUM
890	HAS_DOWN_PAYMENT	Below 2.0%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM

Logical Inconsistency Report

Contradictions detected between triples in the LLM response — independent of the database. These are self-contradictions within the answer itself.

No logical inconsistencies detected.

Symbolic Rule Violations

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

Constraint ID	Rule Type	Severity	Subject	Predicate	Object	Explanation
c_tier_580_619			579	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	subprime	The subject (credit score) 579 is not between 580 and 619 inclusive, violating the constraint for 'subprime' risk tier.
c_interest_rate_range			890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	Below 4.36% (estimated)	The object (interest rate percentage) 'Below 4.36%' cannot be numerically compared as it is not a specific value within the range 0 to 30.
c_down_payment_range			890	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Below 2.0% (required)	The object (down payment percentage) 'Below 2.0%' cannot be numerically compared as it is not a specific value within the range 0 to 50.

Detected Hallucinations

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
575	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT	STRICT		1.00	The object '12.21%' matches '12.21' semantically as the percentage symbol is an addition that does not change the value. Evidence: subject: 575, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 12.21
580	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT	STRICT		1.00	The object '11.91%' matches '11.91' semantically as the percentage symbol is an addition that does not change the value. Evidence: subject: 580, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 11.91
579	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	FABRICATED_FACT	SEMANTIC	SUBJECT	0.00	Subject '579' with predicate exists but no evidence in DB candidates; predicate is known but neither subject nor object appear with it.

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	super prime	FABRICATED_FACT	STRICT	SUBJECT	██████ ████ 0.60	Object 'super prime' is known with predicate, but subject '890' does not appear in DB candidates for this predicate. Evidence: subject: 740, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 745, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 750, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 755, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 760, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 765, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 770, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 775, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 780, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 785, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 790, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 795, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 800, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 805, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime; subject: 810, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime

Triple Audit Trail (Original → Canonical → Verdict)

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict (e.g. dropped due to output truncation).

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
575	HAS_INTEREST_RATE	12.21%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT
580	HAS_INTEREST_RATE	11.91%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT
579	HAS_INTEREST_RATE	11.91%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	FABRICATED_FACT
890	HAS_RISK_TIER	super prime	MEDIUM	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	super prime	FABRICATED_FACT

Verdict Legend

Verdict	Description
CORRECT	All three components (subject, predicate, object) verified against DB.
RULE_VIOLATION	One or more components contradict DB evidence.
MULTI_RULE_VIOLATION	Violation spans multiple DB triples simultaneously.
UNVERIFIABLE	No usable DB evidence found for any component.
PREDICATE_MISMATCH	Subject real, but stated predicate is not a valid relation for it.
FABRICATED_FACT	Triple asserts a relationship that does not exist anywhere in DB.